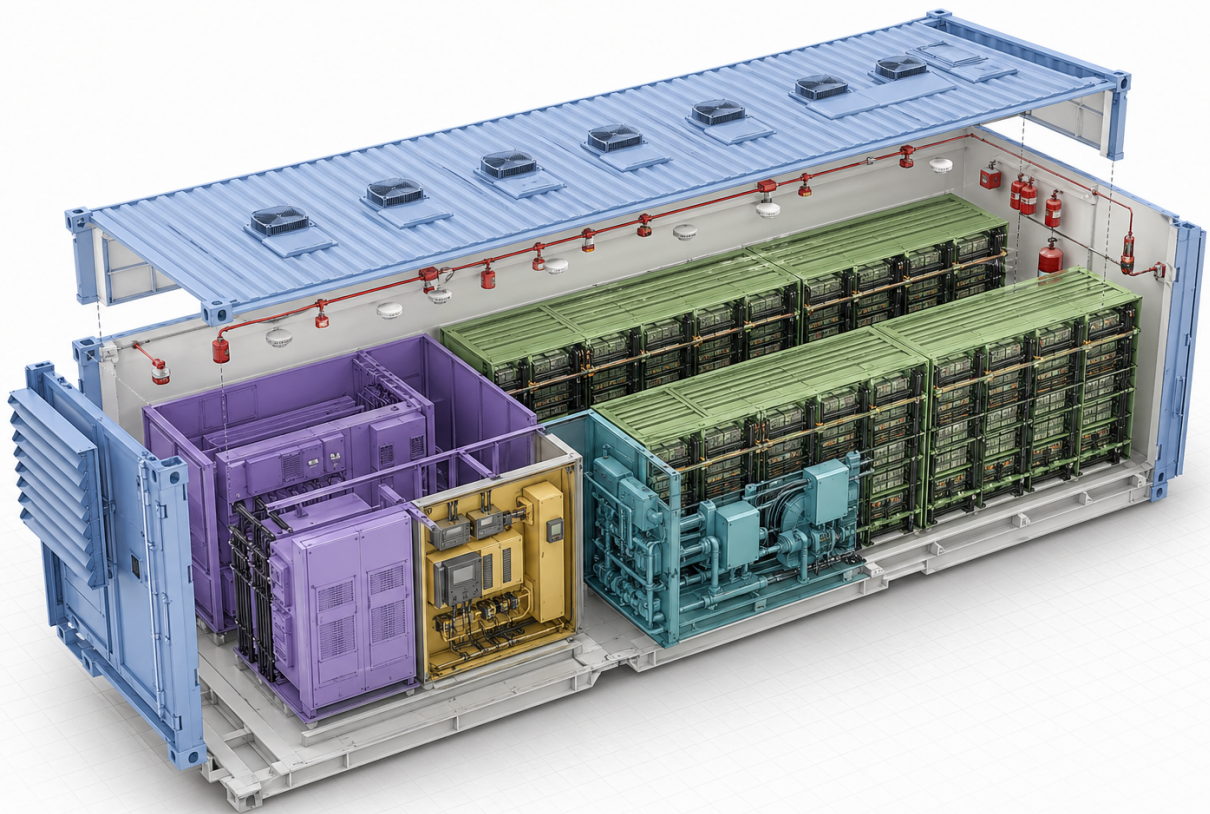


BESS — 40ft 3.5 MWh containerised

Revision A · Fractional Forge · In build

Forge project pack

Project name	BESS — 40ft 3.5 MWh containerised
Revision	Rev A
Project created	25 Apr 2026, 04:42
Brief locked	25 Apr 2026, 06:09
Shipped	Not shipped yet
Document generated	25 Apr 2026, 06:18



Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

MODULES 6	KEY PARTS 43	BOM ROWS 53	SUPPLIERS 29
FAILURE MODES 24	OPEN QUESTIONS 18	STANDARDS 12	REVIEWS 3
UNIT COST £1,276,480	CEILING £180,000	HEADROOM £-1,096,480	

Contents

01	1. Brief
02	2. Regulatory posture
03	3. Sizing optimisation
04	4. Spatial plan
05	5. Modules (one page each)
06	6. BOM master
07	7. Cost waterfall
08	8. Risks register
09	9. Supplier shortlist
10	10. Project audit log

1. Brief

FOUNDER SUBJECT

A 40-foot containerised battery energy storage system delivering 3.5 MWh and 1.5 MW for behind-the-meter commercial and industrial sites in the United Kingdom. Lithium iron phosphate cells, integrated battery management, bi-directional power conversion with grid-forming capability, liquid-cooled thermal management, and certified fire detection and suppression. Target installed capital cost under 180,000 pounds. Must ship complete on a flatbed lorry, install on a prepared concrete pad, and commission inside two days. Compliant with United Kingdom G99 grid code and the latest National Fire Chiefs Council guidance for lithium battery storage.

MISSION

Deliver a factory-integrated 40-foot containerised battery energy storage system providing 3.5 MWh and 1.5 MW of bi-directional power for behind-the-meter commercial and industrial sites in the United Kingdom, commissioning in two days from lorry delivery.

USE CASE

Ship a complete, factory-tested BESS unit on a flatbed lorry, crane it onto a prepared concrete pad, and commission it within two days to reduce peak demand charges and provide grid-forming backup power at a UK C&I site

TARGET CUSTOMERS

UK commercial and industrial sites, DNO-connected behind-the-meter operators, C&I energy buyers subject to demand-charge or grid-constraint pressures

WHY NOW

UK grid code ENA G99 now mandates grid-forming capability for connected assets above 50 kW; NFCC has issued updated lithium battery storage planning guidance creating a compliance moment; combined pressure from rising grid charges and UK net-zero targets is accelerating C&I behind-the-meter storage procurement

Constraints declared

Unit cost ceiling	£180,000
Max mass	55000.00 kg
Target process	not declared
Target material	not declared
Tolerance target	not declared
Quantity target	not declared

2. Regulatory posture

ENA G99 not-started

Distribution network connection — generators and storage above 50 kW

Mandatory UK grid code for behind-the-meter storage connecting at LV/HV; requires grid-forming capability provisions

NFCC BESS Guidance not-started

National Fire Chiefs Council grid-scale battery energy storage planning guidance

UK fire-authority guidance governing siting, separation distances, detection, and suppression design for lithium battery installations

UL 9540A not-started

Large-scale fire propagation test for battery energy storage

Test method validating thermal-runaway propagation behaviour; increasingly required by UK insurers and AHJs for siting approval

UL 9540 not-started

Standard for safety of energy storage systems and equipment

System-level safety standard referenced by authorities having jurisdiction for BESS acceptance

IEC 62619 not-started

Safety requirements for secondary lithium cells and batteries for use in industrial applications

Cell and module-level safety standard for industrial LFP batteries within the racks

IEC 62109-1 not-started

Safety of power converters for use in photovoltaic power systems — Part 1: General requirements

Safety standard for the bi-directional power conversion system

IEC 62109-2 not-started

Safety of power converters — Part 2: Particular requirements for inverters

Inverter-specific safety requirements applied to the grid-forming PCS

NFPA 855 not-started

Standard for the installation of stationary energy storage systems (2023)

Referenced in UK NFCC guidance; governs separation distances, deflagration venting, suppression, and aisle clearances

UL 1973 not-started

Standard for batteries for use in stationary, vehicle auxiliary power and light electric rail applications

Alternative cell and module-level safety standard acceptable in lieu of IEC 62619 for AHJ and insurer acceptance

ISO 668 not-started

Series 1 freight containers — classification, dimensions and ratings

Defines the 40ft High Cube external envelope and corner-casting geometry the container shell must conform to

ISO 1161 not-started

Series 1 freight containers — corner and intermediate fittings

Specifies the forged corner castings used for lifting, lashing, and pad anchoring

ISO 1496

not-started

Series 1 freight containers — specification and testing

Structural performance and test requirements for the ISO container weldment

3. Sizing optimisation

Forge applied the battery_energy_storage v1.0.0 rules library — a closed-form deterministic solve (no trial sweep needed for this domain). Final target below was computed directly from the brief's capacity + envelope constraints.

Envelope

40ft ISO container (standard) · interior 12032×2352×2393mm · 28.30 m² floor

Final config

kw: 100 · kwh: 500 **[FEASIBLE]**

Engineering notes

- Battery density coefficient: 100 kWh/m² — conservative LFP rack density with cable trays + 600mm service aisle.
- Fire agent mass per NFPA 855: 0.04 kg/kWh.

4. Spatial plan — 40ft ISO container (standard)

top_down, plan_type=floor, authored by fang, generated 25 Apr 2026, 06:11 (container_linear v1.0.0)

X — envelope length, mm (0 '12032)

Y — envelope depth, mm (0 '2352)



Floor-mounted Wall-mounted Ceiling-mounted (above) Envelope

Placements

Module	Mount	W×D×H mm	x,y,z mm	Rotation °
Battery Rack Array (LFP)	floor	3600×1200×2200	50,50,0	0
Power Conversion System Bay	floor	625×800×2200	3700,50,0	0
Liquid Cooling & HVAC System	ceiling	2500×600×500	4766,876,1893	0
Fire Detection & Suppression System	ceiling	500×400×300	5766,976,2093	0
BMS & Energy Management Controller	wall	600×300×1800	9582,0,1000	0
ISO Container Shell & Fire-Rated Enclosure	envelope	12032×2352×2393	0,0,0	0

Constraints

separation: ISO Container Shell & Fire-Rated Enclosure "ISO Container Shell & Fire-Rated Enclosure · min 900mm
NFPA 855 §15.3 — minimum separation from fire-suppression nozzle to battery-rack surface.

Notes

- Symmetric layout: no (single-row floor).
- Aisle: 1000mm centred along container depth (2352mm interior).
- Target — KWH: 500 · KW: 100.

3.1 · ISO Container Shell & Fire-Rated Enclosure

Module id: container_shell

6500.00 kg · 14 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Provides the 40ft HC structural envelope, weatherproofing, fire-rated lining, deflagration vents, and ISO corner castings for transport and anchoring.

WHY IT MATTERS

The shell is the single physical asset that makes the system transportable as a complete factory-integrated unit (enabling 2-day commissioning), provides the fire compartmentation that satisfies UK NFCC guidance and insurer requirements, and reacts every internal mechanical and seismic load into the concrete pad. Failure of the shell — structurally, thermally, or through inadequate venting — directly compromises grid-code certification, fire-service acceptance, and the entire commercial proposition.

DESCRIPTION

The container shell is a modified ISO 668-compliant 40ft High Cube weathering-steel (Corten A) container forming the primary structural, environmental, and fire-segregation envelope for the BESS. External dimensions are 12,192 × 2,438 × 2,896 mm with welded corrugated steel sidewalls (1.6–2.0 mm) and a steel pan roof, supported by a perimeter weldment terminated in eight ISO 1161 forged corner castings that serve as the sole lifting, lashing and pad-anchoring interface. Internally, a 75 mm calcium-silicate / mineral-wool lining provides a 120-minute fire rating in line with NFCC guidance and NFPA 855, while low-inertia deflagration vent panels in the roof and end wall manage overpressure from a battery off-gas event. A galvanised unistrut floor grid and reinforced roof frame distribute the concentrated point loads from the battery rack array (10–15 t per row), PCS bay (~2–4 t), and roof-mounted chiller (1.5–2.5 t) into the container's longitudinal bottom rails and corner posts.

KEY PARTS (8)

- ISO 668 40ft High Cube Corten steel container shell, 12,192 × 2,438 × 2,896 mm, 1.6–2.0 mm corrugated side/end walls, welded weathering-steel frame to ISO 1496-1
- ISO 1161 forged steel corner castings, 8 off (4 top, 4 bottom), positioned at 11,985 mm longitudinal / 2,259 mm transverse centres for lifting and anchoring
- Internal fire-rated lining: 75 mm calcium-silicate or mineral-wool panels (e.g. Promat PROMATECT-H equivalent), 120-min fire rating, mechanically fixed to internal frame
- Deflagration vent panels per NFPA 855 §9.6, roof-mounted, low-inertia rupture type (e.g. Fike or Rembe), sized to limit Pred to <100 mbar
- Reinforced floor rails / unistrut grid (P3000-series hot-dip galvanised) bonded to container cross-members for battery rack and PCS anchoring
- Multi-cable transit (MCT) gland plate on door-end wall (e.g. Roxtec RM/CM frame), IP66, fire-rated EI60
- Roof reinforcement frame for chiller/CDU mounting, hot-dip galvanised structural steel, designed for 1,500–2,500 kg distributed roof load
- Foundation anchors: 4 × M30 grade 8.8 chemical anchors (e.g. Hilti HIT-RE 500 V4) into 300–500 mm reinforced concrete pad

FAILURE MODES (4)

- Deflagration vent under-sizing or sticking, causing overpressure rupture of side walls during a thermal-runaway off-gas ignition event
-

Corrosion of Corten shell at cable-gland penetrations and roof-vent flashings, leading to water ingress onto LV busbars and DC arc faults

- Fatigue cracking at corner-casting welds from road transport vibration combined with high payload (45–55 t), risking lifting failure on site
- Fire-rated lining degradation or thermal bridging at penetrations, reducing compartment rating below the 120 min specified and breaching NFCC separation distances

OPEN QUESTIONS (3)

- Confirmed gross transport mass — at the upper end (55 t) the assembly exceeds standard UK 44 t GVW and requires STGO Cat 2 abnormal-load permitting
- Required deflagration vent area, dependent on UL 9540A large-scale fire test results for the chosen cell and module geometry
- Whether AHJ and insurer will accept a single-compartment shell or mandate internal fire-rated segregation between battery zone and PCS bay

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
ISO 668 40ft High Cube Corten steel container shell — Standard container,make UK market price reconditioned		Manual/Assembly · Other	£6,500
ISO 1161 forged steel corner castings (×8) — Standard off-the-shelf con- make tainer fittings		Casting · Other	£960
Internal fire-rated lining (75mm calcium-silicate panels) — Promat or equiv-buy alent, purchased as standard sheets		—	£4,500
Deflagration vent panels (roof-mounted, ×4) — Specialist safety vent, e.g. make Fike, COTS item		Manual/Assembly · Other	£3,200
Reinforced floor rails / unistrut grid (P3000 HDG) — Standard strut chan- buy nels, buy from distributor		—	£1,200
Multi-cable transit gland plate (Roxtec RM/CM) — COTS sealing system, make off-the-shelf		Manual/Assembly · Other	£800
Roof reinforcement frame for chiller mounting — Welded HDG steel frame,make custom fabrication		Welding · Mild Steel	£2,200
Foundation anchors (×4 M30 grade 8.8 chemical anchors) — Hilti HIT-RE buy 500 V4, standard buy		—	£320
Labour	£2,400 — 60 hours at £40/hr for welding, fitting, lining install		
Per-unit total	£21,880		

ASSUMPTIONS

- Reconditioned container price is for one off.
- Fire lining and vents from UK suppliers.

Container shell is buy, internal fire lining and vents are bought, roof frame made.

3.2 · Battery Rack Array (LFP)

Module id: battery_rack_array

24000.00 kg · 20 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Houses LiFePO₄ cell modules in modular racks arranged in two longitudinal rows, providing 3.5 MWh of DC energy storage.

WHY IT MATTERS

This module embodies ~60–70% of the system bill-of-materials cost and defines the energy capacity, cycle life, round-trip efficiency, and dominant safety risk (thermal runaway). Cell selection, module-to-rack mechanical/thermal design, and BMS integration directly determine whether the unit can hit the £180k cost target, pass UL 9540A propagation testing, and satisfy NFCC siting guidance.

DESCRIPTION

The Battery Rack Array is the energy-storage core of the BESS, comprising approximately 16–32 floor-bolted steel racks arranged in two longitudinal rows along the container side walls with an e800 mm centre service aisle. Each rack stacks 16–20 liquid-cooled LFP modules in series to form a ~1500 V DC string, with strings paralleled at a rack-level DC combiner to deliver the full 3.5 MWh capacity to the PCS. Cells are grade-A prismatic LiFePO₄ (280–314 Ah class) selected for cycle life (>6,000 cycles to 80% SoH), thermal stability, and IEC 62619 / UL 1973 certification. Modules sit on aluminium cold plates fed by a 50/50 water-glycol loop via dripless quick-connect couplings, ensuring cell-to-cell $T < 5$ K under 0.43C operation.

KEY PARTS (8)

- LiFePO₄ prismatic cells, 280–314 Ah, 3.2 V nominal (e.g. CATL/EVE/REPT grade A), IEC 62619 / UL 1973 certified — approx. 5,500–6,200 cells total for 3.5 MWh
- Battery modules: 1P52S or 1P48S configuration, ~5.0–5.4 kWh per module, integrated cold plate base, module-level fusing and voltage/temperature sense harness
- Battery racks: 600–800 mm W × 800–1,000 mm D × 2,000–2,200 mm H steel weldment, 16–20 modules per rack stacked vertically, ~1,500 V DC string, 16–32 racks total bolted to container floor unistrut
- Rack-level DC combiner with motorised MCCB or pyro-fuse disconnect (e.g. 1500 V DC, 250–400 A), insulation monitoring, and pre-charge contactor
- Slave BMS (BMU) per module + rack-level master BMS with CAN/Modbus uplink to system EMS
- Aluminium liquid cooling cold plates with quick-connect dripless couplings (e.g. Stäubli CBI/CBC series), EPDM hose runs to ceiling manifold
- Top-of-rack busbar interconnect: tin-plated copper, 1500 V DC rated, with insulating boots and torque-marked bolted joints
- Off-gas sensor head per rack (H₂/CO/VOC, e.g. Nexceris Li-ion Tamer) and module-level smoke aspiration ports

FAILURE MODES (4)

- Single-cell thermal runaway propagating module-to-module due to inadequate cold-plate contact or insufficient inter-module fire barriers
- DC busbar joint loosening from thermal cycling causing localised I²R hotspots and arc-flash risk
- Coolant leak at quick-disconnect couplings causing DC insulation fault and rack trip
- Cell capacity divergence (top-balancing drift) reducing usable energy below 3.5 MWh contractual threshold within 2–3 years

OPEN QUESTIONS (3)

- Final cell supplier and exact cell count — drives rack quantity, total mass, and whether 44 t road limit is achievable

- Rack DC voltage architecture (1500 V single string vs. lower-voltage paralleled) — affects PCS selection and isolation strategy
- Required UL 9540A test configuration (unit-level vs. installation-level) and whether inter-rack fire barriers are needed to pass

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
LiFePO4 prismatic cells 280 Ah (×5500) — Bulk pricing £84/cell, COTS cells, IEC certified	buy	—	£462,000
Battery module assembly (pre-made 1P52S unit) — Assembled module with BMS, buy from integrator	buy	—	£140,400
Battery rack steel weldment (×16) — Custom welded steel rack, local fabrication	make	Welding · Mild Steel	£48,000
Rack-level DC combiner with MCCB (×16) — 1500V DC rated combiner, COTS electrical component	buy	—	£48,000
Slave BMS per module + rack master BMS — Standard BMS boards, off-the-shelf system	buy	—	£32,000
Aluminium cold plates with quick-connect couplings (×320) — COTS cooling plates, standard component	buy	—	£40,000
Top-of-rack busbar interconnect (tin-plated copper) (×16) — Pre-fabricated busbar, buy from electrical supplier	buy	—	£12,000
Off-gas sensor head per rack (×16) + smoke aspiration ports — Nexceris Li-ion Tamer, specialist COTS	buy	—	£32,000
Labour	£3,200 — 80 hours at £40/hr for rack assembly, wiring, plumbing		
Per-unit total	£817,600		

ASSUMPTIONS

- Battery cells price based on 2026 spot market.
- Racks fabricated locally in UK.

Majority buy: cells, modules, BMS, cooling parts; racks are custom make.

ENGINEERING REVIEW (1)

Manufacturing review · warn · 25 Apr 2026, 06:18

The rack array as scoped will not ship complete on a single flatbed inside the United Kingdom 44-tonne road limit, and the module brief is missing every manufacturing parameter (process, material grade, tolerance class, quantity). Production is where the margin lives or dies — and right now we don't have the inputs to cost the line, let alone build it.

[CRITICAL] Transport feasibility: At 24 racks × ~980 kg loaded (18 modules × ~50 kg + ~80 kg weldment), the battery array alone is ~23.5 t. Add container shell (3.8 t), Power Conversion System (2 t), Heating/Ventilation/Air Conditioning chiller (1.5 t), and fire/aux (0.5 t) and gross sits at ~31 t. United Kingdom 6-axle articulated lorry payload is ~29 t. **You are 2–3 t over the road limit before tolerances, cabling, and packaging.** The brief says "ship complete on a flatbed lorry" — that requirement is currently failing arithmetic, not engineering. *Suggestion: Three options, in order of preference: (1) ship the racks empty and install modules on site (kills the 2-day commissioning target — bad); (2) reduce to 20 racks with higher-density 314 Ah cells, drops to ~19.6 t battery + ~7.8 t balance-of-plant = 27.4 t gross, fits with margin; (3) split into 1.75 MWh half-units shipped on two lorries (doubles logistics cost). Lock the cell choice this week — *this is the decision that gates everything downstream*.*

[CRITICAL] Module brief is empty on manufacturing parameters: Target process, target material, tolerance target, and quantity target are all blank. We cannot do a real Design for Manufacturing review against blanks. The rack weldment, cold plate, busbar, and combiner each need their own specification. *Suggestion: Fill the four fields per sub-assembly before tooling discussions. I can produce a draft specification matrix — see recommendation 1.*

[CRITICAL] Single-source risk on cells, couplings, and off-gas sensors: The brief lists CATL/EVE/REPT as candidates (good — three options), but Stäubli dripless couplings and Nexceris Li-ion Tamer are named as if single-sourced. Never single-source anything critical. The day Stäubli's lead time goes to 40 weeks, your 20-week module lead time goes to 40. *Suggestion: Qualify a second coupling source (Colder Products / CPC LQ6 series, or Walther Präzision) and a second off-gas sensor (Honeywell Sensepoint XCD with hydrogen head, or SafeIO). Chase should own this — I'll flag it to him.*

[WARNING] Floor loading on container base: ~1,556 kg/m² under each loaded rack. Standard 40-foot High Cube container floor (28 mm marine plywood on cross-members at ~300 mm centres) is rated to ~1,800 kg/m² distributed but only ~3,000 kg point load. With 24 racks at ~10 kN each, you need a continuous welded steel sub-floor or unistrut grid spreading load to the cross-members — not direct-bolt to the plywood. *Suggestion: Specify a 6 mm steel deck plate welded to the container's longitudinal cross-members, with M10 captive nuts at rack-foot positions. This is also your seismic anchor path.*

[WARNING] Busbar joint integrity vs. thermal cycling — known failure mode you flagged yourself: Tin-plated copper busbars at 1500 V DC carrying 250–400 A will see joint creep under thermal cycling. Torque-marked bolts are necessary but not sufficient. *Suggestion: Specify Belleville (disc spring) washers under each M10 busbar bolt to maintain preload through copper creep. Add an annual infrared thermography inspection requirement to the operations & maintenance manual. Specify joint compound (NO-OX-ID A-Special or equivalent) — without it, oxide growth at the joint interface will raise resistance over 5+ years. *If we fix this one thing, three downstream problems disappear* — hotspot risk, arc-flash risk, and capacity-fade-from-impedanc*

[WARNING] Cold plate flatness and cell-to-plate contact pressure unspecified: Cell-to-cell T < 5 K under 0.43C is a real number, but it depends entirely on cold-plate flatness (typically 0.1 mm/m) and thermal interface material thickness/compression. Aluminium cold plates produced by friction-stir-welded extrusion with internal channels are the production-proven path; brazed plates are cheaper at low volume but warp. *Suggestion: Specify cold-plate flatness d 0.15 mm over the cell footprint, thermal interface material (TIM) compression range 20–40%, and cell-pack pressure 100–300 kPa per cell manufacturer datasheet. Add a flatness inspection at incoming-goods.*

[WARNING] Rack weldment idiot index acceptable, but inspection is the risk: Raw steel ~£96/rack, finished welded rack ~£600 'idiot index ~6.3, which is green (target <10). The cost is in welding and post-weld straightening, not material. *What does the process look like on paper versus what actually happens?* On paper, robotic Metal Inert Gas welding holds ±2 mm over a 2 m frame. In reality, distortion eats half your tolerance budget unless you weld in a fixture and stress-relieve. *Suggestion: Specify weld fixture with hard datums at the four module-mounting rails. Require post-weld dimensional inspection at four points per rack (corner-to-corner diagonals). Reject if diagonals differ by >3 mm — the modules won't slide in cleanly otherwise.*

[INFO] Tolerance class not specified: For a steel weldment of this size, ISO 13920 class B (general welded structures) is the right target — ±2 mm linear, ±45' angular over 2 m. ISO 2768-mK is for machined parts and is the wrong reference for a welded rack. Don't over-specify.

[INFO] DC architecture decision still open: 1500 V single-string vs. paralleled lower-voltage strings is listed as an open question. This is a Power Conversion System interface decision, not a rack manufacturing decision per se, but it changes the rack-level combiner specification (250 A vs. 400 A pyro-fuse, single-pole vs. double-pole isolation). Lock this before ordering long-lead combiner hardware (16-week lead).

RECOMMENDATIONS

- ****Fill the manufacturing specification matrix this week.**** Four sub-assemblies, four fields each. I can draft this — want me to produce it as a deliverable you can drop straight into the Specify page?
- ****Resolve the mass/transport problem before any other work.**** Cell selection drives everything — rack count, gross mass, container floor design, anchor pattern. *Run The Algorithm on that part before we talk tooling — question the requirement first.* Step 1: does it really need to be 3.5 megawatt-hours nameplate, or 3.5 megawatt-hours usable at end-of-warranty? Those are different boundary conditions and one of them lets you ship.

- ****Dual-source the four single-source risks**** (cells, dripless couplings, off-gas sensors, pyro-fuse disconnects). Hand to Chase with a 6-week qualification deadline.
- ****Default to contract manufacturing for the rack weldments.**** *Don't buy the machine shop. Find a better contract manufacturer and design a cleaner part.* United Kingdom-based shops that do battery rack weldments at 100–500 unit/year volume: G&O Springs (Birmingham), Bri-Stor Systems (Stafford), Mec Com (Coventry). Get three first-article quotes against the same drawing pack — *first article in 72 hours or we're not trying hard enough* once the drawings are released.
- ****Specify the rack/module mating interface as the critical assembly tolerance.**** Module slides on cold-plate rails with dripless couplings engaging axially. Worst-case stack of 18 modules in a 2 m rack with ± 0.3 mm per pocket = ± 5.4 mm worst case, ± 1.3 mm root-sum-square. Couplings need ± 2 mm float — design the module bracket with slotted holes, not round, on the rear support.
- ****ASSEMBLY: Sequence — assemble the racks LATE.**** Order: (1) container shell + sub-floor + cable trays, (2) Heating-Ventilation-Air-Conditioning ducts and coolant manifold ceiling runs, (3) Power Conversion System bay, (4) empty racks bolted to sub-floor, (5) modules slotted into racks, (6) busbars and combiners, (7) Battery Management System harness, (8) fire suppression. Modules go in last because they are the heaviest, most expensive, and most damage-sensitive items — minimise time on the line.
- ****ASSEMBLY: Fixtures required.**** (a) Rack-positioning template that locates all 24 racks to ± 2 mm against the container datum corner before bolting. (b) Module insertion trolley with hydraulic lift (50 kg modules at 1.8 m height is a manual handling failure waiting to happen — Health and Safety Executive will fail the assembly cell audit). (c) Busbar torque-and-mark station with calibrated digital torque wrench, paint pen, and inspection log.
- ****ASSEMBLY: Critical assembly tolerances.**** Rack vertical plumb ± 2 mm over 2 m. Inter-rack pitch ± 3 mm. Cold-plate coolant port axial position ± 1.5 mm to mate with ceiling manifold drop. Busbar bolt torque 45 Nm $\pm 10\%$ with paint witness mark.
- ****ASSEMBLY: Test points after this module is built.**** (1) Hi-pot test rack-to-chassis at 3 kV DC for 60 seconds before any cells installed. (2) Insulation Resistance ≥ 100 M Ω at 1000 V DC. (3) Coolant pressure test at 1.5 $\times$ working pressure for 30 minutes — zero drips. (4) Battery Management System communication handshake on every module before sealing the rack covers. (5) Off-gas sensor calibration check with hydrogen test gas. *Do not* energise cells until 1–4 pass.
- ****ASSEMBLY: Dependencies.**** Container shell + sub-floor must be complete and dimensionally inspected. Cooling manifold ceiling drops must be installed and pressure-tested. Cable trays must be in place. Power Conversion System bay floor anchors set. The rack array is the densest, heaviest sub-system in the box — everything that can't be installed afterwards must be installed before.

3.3 · Power Conversion System Bay

Module id: pcs_bay

3500.00 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Bi-directional grid-forming PCS converting between battery DC and AC grid power at 1.5 MW, including AC switchgear and cable gland plate to external transformer.

WHY IT MATTERS

The PCS Bay is the sole electrical interface between the stored DC energy and the customer/grid AC network; it dictates compliance with G99 grid code, dispatch performance, round-trip efficiency, and grid-forming behaviour. Its segregation and protection design also define the system's fault-clearing performance and fire-isolation strategy, both of which are gating items for DNO acceptance and insurer approval.

DESCRIPTION

The PCS Bay is a segregated compartment at the door end of the 40ft container, approximately 2,000 mm long × 2,350 mm wide × 2,698 mm high, housing the bi-directional grid-forming inverter stack, AC switchgear, DC protection, auxiliary distribution, and external cable gland plate. The bay is isolated from the battery compartment by a 60-minute fire-rated partition with IP-sealed penetrations for DC busbars, coolant pipework, and control cabling. The PCS itself is a modular liquid-cooled IGBT or SiC-based inverter (typically 4–6 parallel power blocks) converting 600–900 VDC battery-side to 400 VAC three-phase grid-side at 1.5 MW continuous, with grid-forming firmware enabling voltage/frequency reference, virtual inertia, and black-start per evolving G99 / G100 provisions.

KEY PARTS (7)

- Bi-directional grid-forming PCS, 1.5 MW continuous, modular (4× 375 kW or 6× 250 kW IGBT/SiC inverter blocks), liquid-cooled cold plates, 600–900 VDC battery side / 400 VAC 3-phase grid side, IEC 62109-1/-2 and G99-compliant, grid-forming firmware with black-start capability
- AC switchgear cabinet — 2000 A moulded-case circuit breaker with motorised operator, 400 V 3-phase 4-wire busbar, surge protection (Type 1+2 SPD), earth-fault and arc-flash relay, IEC 61439-2 form 3b assembly
- DC isolation & protection panel — 1500 VDC-rated fused disconnects per battery string, pre-charge circuit, DC contactors (e1500 A), insulation monitoring device (IMR) per IEC 61557-8
- Cable gland plate — 6 mm galvanised steel, IP66 multi-cable transit (e.g. Roxtec RM/B) for 3× 240 mm² AC power cables, control/fibre, and earth bonding to external transformer
- Internal liquid cooling manifold — stainless steel, 1" supply/return with quick-disconnect couplings to PCS cold plates, 50/50 water-glycol, 30–50 L/min nominal flow
- Auxiliary LV distribution panel — 230 V AC site supply, 24 VDC control, UPS-backed for BMS/EMS, fire panel, and emergency lighting
- Bay segregation wall — 60-minute fire-rated partition (calcium-silicate, 60 mm) with sealed penetrations isolating PCS bay from battery compartment per NFPA 855 §9.6.5

FAILURE MODES (4)

- IGBT/SiC module failure due to thermal cycling or coolant flow loss, causing inverter trip and loss of dispatch
- DC bus insulation degradation or earth fault leading to IMR alarm and forced shutdown
- AC breaker fails to clear a downstream fault, causing arc-flash event within the bay
- Coolant leak from manifold or cold plate onto live DC busbars, risking short-circuit and fire propagation

OPEN QUESTIONS (3)

- Final PCS vendor selection (e.g. Sungrow, SMA, Power Electronics, EPC) and whether 4× 375 kW or 6× 250 kW topology is preferred for redundancy vs. cost
- Whether the LV/MV transformer is supplied within the container envelope or as an external skid — affects gland plate sizing and AC voltage class
- Confirmation that grid-forming firmware meets the specific DNO's interpretation of G99 / G100 at the time of commissioning

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Bi-directional grid-forming PCS (1.5 MW modular) — Large PCS, COTS from OEM like SMA/Huawei	buy	—	£265,000
AC switchgear cabinet (2000A MCCB, SPD, arc-flash relay) — Standard switchgear assembly, off-the-shelf	buy	—	£25,000
DC isolation & protection panel (1500VDC fused disconnects) — COTS DCmake protection panel with IMR		Manual/Assembly · Other	£18,000
Cable gland plate (6mm galvanized steel) with Roxtec frame — Standard gland plate and Roxtec seal	make	Plating · Zinc	£600
Internal liquid cooling manifold (stainless steel) — Custom welded manifold with quick-connects	make	Welding · Stainless Steel	£1,500
Auxiliary LV distribution panel (230VAC/24VDC, UPS-backed) — Standard LV panel with UPS, COTS	make	Manual/Assembly · Other	£4,000
Bay segregation wall (60-min fire-rated calcium-silicate) — Prefabricated fire panels, buy from supplier	buy	—	£1,200
Labour	£1,600 — 40 hours at £40/hr for installation of panels and piping		
Per-unit total	£316,900		

ASSUMPTIONS

- PCS price includes integrated cooling controls.
- All panels standard sizes from UK distributors.

Almost all components are buy; only cooling manifold is custom make.

3.4 · Liquid Cooling & HVAC System

Module id: thermal_management

1450.00 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Roof/end-wall mounted chiller, coolant manifolds, and internal HVAC providing temperature regulation to battery racks, PCS, and ambient air space.

WHY IT MATTERS

LFP cells must be held within 15–35 °C for rated cycle life; thermal gradients above 5 °C between cells accelerate ageing and trigger BMS derating, directly limiting throughput and warranty validity. The cooling system also evacuates >100 kW of waste heat at full 1.5 MW operation — without it, the container would thermally runaway in minutes. Cooling availability gates the entire system's power rating and is a prerequisite for fire safety (excess heat is the precursor to thermal runaway).

DESCRIPTION

The thermal management module is a closed-loop water-glycol (50/50 propylene glycol) liquid cooling system combined with a smaller air-conditioning unit for ambient space conditioning. A roof-mounted packaged chiller (80–100 kW capacity) rejects heat to ambient via EC-fan-driven microchannel condenser coils, and circulates ~30–40 °C coolant through a Coolant Distribution Unit (CDU) housing redundant pumps, expansion tank, filtration, and instrumentation. Coolant is distributed via a stainless steel 316L ring main running along the container ceiling, with branch drops feeding cold plates inside each battery rack and a dedicated loop to the PCS bay; quick-disconnect dripless couplings at each rack permit module hot-swap during service. A separate end-wall HVAC unit (12–18 kW) maintains aisle and PCS bay air temperature/humidity to prevent condensation on busbars and electronics.

KEY PARTS (6)

- Roof-mounted packaged chiller, 80–100 kW nominal cooling capacity, R454B or R513A refrigerant, scroll compressor with EC condenser fans, IP55 enclosure (~1,800 × 1,200 × 1,500 mm)
- Coolant Distribution Unit (CDU) with redundant 50/50 water-glycol pumps (2 × 100% duty/standby), expansion tank, 50 µm filtration, and pressure/flow sensors; stainless steel manifold rated 6 bar
- Stainless steel 316L coolant ring main with EPDM-lined flexible jumpers to each battery rack, quick-disconnect dripless couplings (Stäubli CBI or equivalent) at rack interface
- Internal HVAC unit (12–18 kW sensible cooling) with electric reheat and dehumidification, end-wall mounted, ducted along ceiling for ambient air conditioning of PCS bay and aisle
- PT1000 temperature sensors (±0.3 °C) and ultrasonic flow meters per cooling zone; Modbus TCP gateway to EMS
- Glycol leak detection rope sensors routed under manifolds and along container floor sump

FAILURE MODES (4)

- Glycol leak at rack jumper coupling causing low-flow alarm, BMS derate, and potential DC arc-fault if coolant contacts busbars
- Compressor failure or refrigerant loss in chiller leading to total loss of battery cooling and forced system shutdown
- Pump bearing wear or controller fault on duty pump without successful failover to standby, causing localised hot-spots
- Condensation on cold coolant lines or cold plates in humid UK summer conditions, dripping onto electrical equipment

OPEN QUESTIONS (3)

- Final heat load split between battery racks (continuous ~60–80 kW) and PCS (peak ~30–45 kW) pending PCS efficiency curve confirmation
- Whether a single chiller is acceptable or N+1 redundancy is required by insurer / customer SLA
- Acoustic emission limits at site boundary — may drive condenser fan selection and night-mode operation logic

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Roof-mounted packaged chiller — COTS industrial chiller, RS/direct supplier price.	buy	—	£25,000
Coolant Distribution Unit (CDU) with pumps, tank, manifold — Standard CDU from thermal management suppliers (e.g., Boyd).	make	Manual/Assembly · Other	£12,000
Stainless steel 316L coolant ring main — Custom pipe fabrication, welded 316L, per design size.	make	Welding · Stainless Steel 316L	£3,500
EPDM-lined flexible jumpers with quick-disconnect couplings (×12) — COTS Stäubli CBI couplings and hose assemblies.	buy	—	£1,800
Internal HVAC unit (12-18 kW) — Standard ducted HVAC unit from major brands.	buy	—	£4,500
PT1000 temperature sensors (×6) — COTS sensors from RS Components, £50 each.	buy	—	£300
Ultrasonic flow meters (×6) — COTS ultrasonic flow meters, £200 each.	buy	—	£1,200
Modbus TCP gateway — Standard industrial gateway, RS price.	buy	—	£400
Glycol leak detection rope sensors and controller — COTS leak detection system from TraceTek or similar.	buy	—	£1,200
Labour	£2,400 — 60 hrs at £40/hr for pipe fitting, wiring, installation.		
Per-unit total	£52,300		

ASSUMPTIONS

- Chiller and CDU are COTS industrial units.
- Simplified installation labour for a single container.
- Price excludes shipping and VAT.

Majority buy items with custom welded ring main.

ENGINEERING REVIEW (1)

Manufacturing review · warn · 25 Apr 2026, 06:18

The architecture is sound and the manifold is structurally over-engineered (good for safety, expensive for production), but the chiller is ****undersized for combined peak load****, single-source risk on the chiller is unmitigated, and the stainless ring main has an idiot index of 10–14x — fabrication labour is eating the margin.

[CRITICAL] Thermal capacity: Combined continuous load (battery ~70 kW + Power Conversion System ~20 kW = 90 kW) ****exceeds 80 kW nominal chiller capacity by 13%****. Peak combined load reaches ~105 kW. At a UK summer ambient of 32°C the chiller will already be derated by 10–15% from its nameplate. You will hit thermal alarm and Battery Management System derate on hot days during peak dispatch. *Suggestion: Specify a 120 kW nominal chiller as the baseline, or split into 2 × 60 kW units for N+1 redundancy (also resolves Open Question #2). The cost delta from 80 kW ' 120 kW is roughly £4–6k on a £25–35k unit — trivial against the consequence of summer derate.*

[CRITICAL] Single-source chiller, 16-week lead time: One packaged chiller from one supplier is your single biggest schedule and warranty risk. If that supplier slips, the entire 40-foot Battery Energy Storage System ships late. Never single-source anything critical — I don't care if it costs 15% more, the day your only supplier fails, your company fails. *Suggestion: Qualify two chiller suppliers in parallel (Carrier AquaSnap, Trane, Daikin, Mitsubishi Heavy Industries are all credible at this capacity). Spec to a generic performance envelope — coolant supply temperature, flow rate, refrigerant family, electrical interface, mounting footprint — not a part number.*

[CRITICAL] Refrigerant choice locks you to F-Gas regulation timeline: R454B (GWP 466) is the right call for new United Kingdom installations under the F-Gas phase-down. R513A (GWP 631) is being squeezed and may be hard to service in 5 years. Pick one and commit. *Suggestion: Standardise on R454B. It's mildly flammable (A2L) — confirm the chiller is on the roof outside the fire-rated envelope so the A2L charge limit inside the container is not relevant.*

[WARNING] Idiot index on stainless manifold = 10–14x: Raw 316L for ~30 m of DN65 ring main is about £840. Finished welded, pickled, passivated, pressure-tested assembly quotes at £8–12k. The fabrication labour (orbital welding, internal cleanliness, hydrostatic test, certification) is doing all the work. High idiot index means the design is the problem, not the supplier. *Suggestion: Run The Algorithm on the manifold before quoting. Step 1, question the requirement: does it need to be 316L, or will 304L do? You're running propylene glycol at 30–40°C, not seawater — 304L cuts material cost ~25% and welds cleaner. Step 2, delete: do you need a full ring main, or will a single-ended header with isolation valves work? A ring gives redundancy on paper but doubles your weld count. Step 3, simplify: pre-fabricated modular skid suppliers (HRS Heat Exchangers, Klim*

[WARNING] Quick-disconnect couplings are a leak source by design: Stäubli CBI dripless couplings are the right specification, but every coupling is a potential leak point that can drip onto Direct Current busbars. Listed failure mode #1 confirms this. Industry yield experience: roughly 1 in 500 couplings will weep within the first year if installed by a non-specialist. *Suggestion: Specify factory-installed couplings only — not field-installed. Pressure-test every coupling at 1.5× design pressure for 30 minutes before the rack closes up. Add a drip tray under each coupling, ducted to the floor sump where the leak rope sensor is. **If we fix this one thing, three downstream problems disappear** — leak detection latency, busbar contamination risk, and warranty exposure on cell modules.*

[WARNING] Condensation on cold lines (Failure Mode #4): UK summer dewpoint regularly hits 18–20°C. If your coolant supply is 25°C and your aisle Heating, Ventilation, and Air Conditioning unit fails or under-performs, you will get condensation on the cold plates and manifolds. This is a known kill mode for liquid-cooled BESS. *Suggestion: Two locks: (1) closed-cell elastomeric pipe insulation (Armaflex Class 0 or equivalent, 19 mm wall) on every coolant line inside the container — non-negotiable, mandate at the manifold supplier; (2) interlock the coolant supply temperature setpoint to the aisle dewpoint sensor. If aisle dewpoint > coolant supply minus 3°C, raise the supply temperature setpoint.*

[WARNING] PT1000 ±0.3°C is tighter than you need: ±0.3°C is precision-grade. For battery thermal control you make decisions on 2–5°C bands. Over-tolerancing the sensor doesn't make the control loop better — it just costs more and adds calibration burden. *Suggestion: Drop to Class B PT1000 (±0.55°C at 25°C, achievable with stock parts at 1/3 the unit cost). Reserve ±0.3°C only for the chiller supply/return sensors used for performance acceptance test.*

[INFO] Manifold hoop stress is fine: DN65 schedule 10S 316L at 6 bar design pressure runs a safety factor of 17x against yield. Structurally over-engineered, which is correct for a pressure system in a battery enclosure. No change needed — the cost issue is fabrication, not material. *Suggestion: Hand off to Jian for the roof sub-frame load path. He needs your final chiller mass + footprint + fan-induced dynamic loads by end of week to size the sub-frame inside the shell tooling lead time.*

[INFO] Roof loading — coordinate with shell module: A packaged 80–120 kW chiller weighs 800–1,200 kg. The ISO container roof is not designed to carry that as a distributed load. You need a structural sub-frame welded to the corner posts, not the roof skin. *Suggestion: Hand off to Jian for the roof sub-frame load path. He needs your final chiller mass + footprint + fan-induced dynamic loads by end of week to size the sub-frame inside the shell tooling lead time.*

RECOMMENDATIONS

- ****Owner: Lead Engineer. By Friday.**** Re-spec the chiller envelope: 120 kW nominal (or 2 × 60 kW N+1), R454B refrigerant, generic performance specification not a part number, two qualified suppliers.
-

****Owner: Lead Engineer. Next week.**** Run The Algorithm on the manifold: question 316L vs 304L, question ring vs header topology, delete every weld you can't justify. Target idiot index d6x.

- ****Owner: Manufacturing (me). Two weeks.**** Get quotes from three pre-fabricated coolant distribution unit suppliers (HRS Heat Exchangers, Klima-Therm, an equivalent United Kingdom fabricator) for a skid-mounted assembly. ****First article in 72 hours from supplier selection or we're not trying hard enough.**** Compare against a build-to-print quote from a stainless fabricator.
- ****Owner: Lead Engineer + Chase.**** Lock refrigerant on R454B across all chiller suppliers in the request for quotation. F-Gas compliance is non-negotiable for United Kingdom installation.
- ****Owner: Manufacturing.**** Mandate factory-installed and factory-tested quick-disconnect couplings. Write the acceptance test protocol: 1.5× design pressure, 30 minutes, zero pressure decay, dye penetrant on every weld.
- ****Owner: Controls.**** Implement dewpoint-interlocked coolant supply temperature setpoint. This is a software change, costs nothing, prevents a £100k warranty event.
- ****Owner: Lead Engineer.**** Drop sensor specification to Class B PT1000 except at chiller acceptance test points. Save the budget for redundant flow meters.
- ****ASSEMBLY: Sequence — middle/late.**** Install after battery rack array is mechanically placed but before electrical commissioning. The manifold ring main mounts to the ceiling, then drops to each rack — the racks must be in their final position to set drop lengths.
- ****ASSEMBLY: Fixtures.**** You need (a) a ceiling-mount manifold cradle jig that holds the ring main at correct height and slope (1:200 fall to drain point) during welding/clamping, (b) a torque-controlled tool kit for the Stäubli couplings — they are torque-sensitive, and (c) a hydrostatic test rig with calibrated pressure gauge and chart recorder for the acceptance test.
- ****ASSEMBLY: Critical tolerances.**** Manifold drop centreline to rack inlet centreline: ± 3 mm — this is what makes the flexible Ethylene Propylene Diene Monomer jumpers fit without strain. Slope of ring main: minimum 1:200 fall to drain point — flat sections trap air. Coupling face cleanliness: zero contamination, factory-bagged until mating.
- ****ASSEMBLY: Test points (verify before closing the ceiling).**** (a) Hydrostatic test at 1.5× design pressure for 30 minutes, zero decay; (b) flush-and-fill cleanliness — particle count <NAS 8 on return filter after 1 hour circulation; (c) flow balance per rack within $\pm 10\%$ of design; (d) leak rope sensor wet test on every zone.
- ****ASSEMBLY: Dependencies.**** Must follow: ISO Container Shell (anchor points and roof sub-frame), Battery Rack Array (drop length geometry). Must precede: Power Conversion System Bay electrical commissioning (you cannot energise busbars until coolant lines are pressure-tested and insulated), Fire Detection and Suppression System final commissioning (suppression nozzle layout depends on final pipe routing).

3.5 · BMS & Energy Management Controller

Module id: bms_ems_controller

120.00 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Master controller integrating per-rack BMS, energy management, grid-code logic, and supervisory control of PCS, thermal, and fire systems.

WHY IT MATTERS

This module enforces cell-level safety (over-voltage, over-temperature, isolation faults), executes grid-forming and grid-following dispatch in compliance with G99, coordinates thermal setpoints, and provides the hardwired interlock chain that trips contactors and releases suppression on fire/off-gas alarms. Without it the system cannot safely charge, discharge, or respond to fault conditions, and cannot achieve DNO acceptance.

DESCRIPTION

The BMS & EMS Controller module is the digital nervous system of the BESS — a wall-mounted IP54 control cabinet (H800 × 2,000 × 600 mm) located in the PCS bay at the door end of the container, housing a master BMS controller, an industrial PLC-based EMS, a redundant Ethernet ring switch, 24 VDC UPS-backed auxiliary supply, and SIL-rated hardwired safety I/O. Per-rack slave BMS PCBAs are distributed inside each battery rack and communicate to the master via isolated CAN bus over shielded twisted-pair, while the EMS interfaces northbound to the site SCADA/DNO interface via IEC 61850, DNP3, or Modbus TCP and southbound to the PCS, chiller, and fire panel via Modbus RTU/TCP and hardwired interlocks.

KEY PARTS (7)

- Master BMS controller unit (e.g. equivalent to Nuvation G5 or Orion BMS stack-level), 19" rack-mount 2U, CAN/Modbus TCP, supporting up to 32 slave BMS nodes and 1500 VDC string monitoring
- Per-rack slave BMS modules (16–32 off), measuring cell voltage (± 2 mV accuracy), cell temperature (± 1 °C, NTC sensors), and providing passive balancing at 100–200 mA per cell
- Energy Management System (EMS) industrial PC / PLC — e.g. Beckhoff CX-series or Siemens S7-1500, with IEC 61850 and DNP3 support for G99 grid-code interfacing
- Managed industrial Ethernet switch, ring-redundant (e.g. Hirschmann RSP series), 8–16 port, DIN-rail mounted, IP30
- 24 VDC auxiliary power supply with UPS backup (e30 min hold-up via VRLA or LFP aux battery) for controllers and safety I/O
- Hardwired safety I/O module (SIL 2) interfacing to PCS trip, contactor open commands, and suppression release relays
- Wall-mounted IP54 control cabinet, ~800 × 2,000 × 600 mm, powder-coated mild steel, housing controllers, switch, PSU, and terminal blocks

FAILURE MODES (4)

- Loss of CAN communication to one or more slave BMS modules causing loss of cell visibility and forced derate or trip
- EMS firmware fault or watchdog hang leading to stale dispatch commands and possible grid-code non-compliance
- Auxiliary 24 VDC supply failure with UPS depletion, causing uncontrolled shutdown of contactors and cooling
- Cyber-intrusion via SCADA/remote-access interface enabling unauthorised dispatch or safety-system bypass

OPEN QUESTIONS (3)

- Final selection of EMS platform and whether grid-forming control logic is hosted on EMS PLC or embedded in PCS firmware
- Required SIL rating for the suppression-release and contactor-trip safety function (SIL 2 vs SIL 3) per insurer/AHJ

- Northbound protocol stack required by target DNOs and aggregator partners (IEC 61850 vs DNP3 vs proprietary)

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Master BMS controller (2U rack mount) — COTS BMS controller, e.g. Nuvation G5 class.	buy	—	£4,000
Per-rack slave BMS modules (×24) — COTS slave modules, ~£500 each for 24 racks.	buy	—	£12,000
EMS industrial PC/PLC (Beckhoff CX-series) — COTS PLC with IEC 61850 from RS/Farnell.	buy	—	£5,000
Managed industrial Ethernet switch (8-port) — COTS Hirschmann RSP switch from RS.	buy	—	£800
24 VDC PSU with UPS backup (30 min hold-up) — COTS DC UPS system from Phoenix Contact.	buy	—	£1,200
Hardwired safety I/O module (SIL 2) — COTS safety module from Siemens (ET200SP).	buy	—	£1,500
Wall-mounted IP54 control cabinet (800x2000x600 mm) — Standard enclosure from RS, painted mild steel.	buy	—	£600
Labour	£2,400 — 60 hrs at £40/hr for wiring, mounting, programming.		
Per-unit total	£27,500		

ASSUMPTIONS

- All controllers and modules are COTS.
- Labour includes wiring and PLC programming.
- Prices based on standard industrial suppliers.

All parts are standard buy items; no custom fabrication.

ENGINEERING REVIEW (1)

Manufacturing review · warn · 25 Apr 2026, 06:18

The module is fundamentally sound as a system-integration exercise — it's mostly bought-in industrial kit in a sheet-metal cabinet — but there are real manufacturability and assembly risks around the IP54 cabinet thermal load, sheet-metal tolerances around door seals, and a single-source/cyber risk pattern that will bite you in the field.

[CRITICAL] Thermal — IP54 sealed cabinet with continuous controller load: An 800 × 2,000 × 600 mm steel cabinet with a Beckhoff/Siemens PLC, managed switch, 24 VDC PSU, UPS battery, and SIL safety I/O will dissipate 200–400 W continuously. Quick hand calc: ~3.92 m² external surface, natural convection h ~6–8 W/m²K through painted steel, container internal ambient regularly hits 40–45 °C in UK summer once you account for PCS and battery waste heat. Steady-state internal rise above container ambient $H/Q / (h \cdot A) = 300 / (7 \times 3.92) \approx 11$ °C above container ambient, before any solar gain on the container shell**. That puts internal cabinet air at 50–55 °C — above the 45–50 °C derating point for most industrial PLCs and Ethernet switches. *Source: hand calculation, h-value from industrial enclosure reference data; flag as estimate.* *Suggestion: Either (a) specify an IP54 cabinet air-conditioner or thermoelectric cooler sized for 400 W at 50 °C ambient (Rittal Blue e+, Pfannenberger DTI/DTS — 600–1200 GBP), or (b) tap the existing liquid-cooling loop with a small air-to-liquid heat exchanger and run the cabinet as a sealed loop. Coordinate with the Liquid Cooling module owner — this is the kind of thing where if we fix this one thing, three downstream problems disappear.*

[CRITICAL] Single-source / cyber risk on EMS platform: Beckhoff CX vs Siemens S7-1500 is a binary choice in your spec, and the open question on northbound protocol stack means you may end up locked into one vendor's IEC 61850 stack. Combined

with the listed cyber-intrusion failure mode, this is a structural risk. Never single-source anything critical. The day Siemens has a 12-month lead time (which happened in 2022–2023, real lead times of 50+ weeks on S7-1500), your project stops. *Suggestion: Pick an EMS platform that runs on commodity industrial PC hardware (CODESYS-based — Wago, Beckhoff, Phoenix Contact all run it; or a Linux-based EMS like Open Energy Modelling stack). Specify the protocol stack and application logic as portable IEC 61131-3 / IEC 61499 code, not vendor-locked function blocks. Engage Chase to dual-source the controller hardware before tooling the cabinet panel layout.*

[WARNING] Sheet metal — IP54 rating with 800 × 2000 mm door: The cabinet is at the upper edge of the sheet metal process envelope (max 3000 × 1500 × 300 mm per the database — your 600 mm depth is fine, but a 2 m tall door panel will need stiffening to hold flatness for the IP54 gasket seal). Standard sheet-metal tolerance is ±0.1 mm on features but flatness over a 2 m door is typically ±2–3 mm without internal ribs or a formed return flange. IP54 needs gasket compression of 30–50% — flatness deviation > gasket thickness = leak path. *Suggestion: Specify the door as either (a) bought-in from a standard cabinet vendor (Rittal TS 8, Schneider Spacial SF, Eldon MAS) — they've solved this and the cabinet is a commodity at 800–1500 GBP, or (b) if custom, require formed return flanges on all four edges and an internal stiffener channel. Run the algorithm on this part before we talk tooling — question the requirement first. The idiot index on a custom 2 m sheet-steel cabinet is brutal: ~30 kg of mild steel at £2/kg = £60 raw,*

[WARNING] Tolerance specification — none provided: Module brief has no target tolerance, process, material, or quantity. For a controls cabinet that's fine for the cabinet itself (use ISO 2768-m, medium class — covers ±0.5 mm at this size and is free), but the slave BMS PCBA mounting in the battery racks needs explicit tolerance because cell-voltage measurement at ±2 mV requires controlled wiring loom lengths and connector positions. *Suggestion: Specify ISO 2768-m for cabinet sheet metal. Specify ±0.5 mm positional on PCBA mounting holes and DIN-rail location. Specify wiring loom length tolerance ±10 mm for the cell-tap harnesses (this matters for voltage measurement noise floor — talk to Jian).*

[WARNING] CAN bus topology — known failure mode is in your spec: You've already listed "Loss of CAN communication to one or more slave BMS modules" as a known failure mode but the design is a single CAN bus with isolated transceivers. At 32 nodes and 1500 VDC string monitoring, a single shorted stub kills visibility on the whole string. *Suggestion: Dual-redundant CAN bus (CAN-A and CAN-B), or migrate to isolated daisy-chain with bypass (Texas Instruments BQ79616-style architecture used by Tesla, BMW). This is an architecture call — flag to Max.*

[INFO] SIL rating open question: SIL 2 is almost certainly what your insurer and the Distribution Network Operator will accept for contactor trip and suppression release on a behind-the-meter Battery Energy Storage System under National Fire Chiefs Council guidance. SIL 3 doubles the cost of safety I/O modules (Siemens F-series, Beckhoff TwinSAFE) and adds 4–6 weeks to commissioning for proof-test documentation. *Suggestion: Resolve this with the insurer and Authority Having Jurisdiction in writing **before** ordering safety I/O hardware. 16-week lead time means this decision is on the critical path right now. Talk to Leo on the legal/insurer interface.*

[INFO] Material-process compatibility: Powder-coated mild steel for the cabinet is correct — sheet-metal compatible per the database (0.5–6 mm range, your likely 1.5–2 mm wall fits cleanly), and Steel 1018 yield of 370 MPa is dramatically over-spec for a controls cabinet. No issue, just noting it's the right call.

RECOMMENDATIONS

- ****Buy the cabinet, don't build it.**** Specify a Rittal TS 8 or Schneider Spacial SF 800 × 2000 × 600 IP55, modify with cutouts. Saves 4–6 weeks of sheet-metal lead time and removes the IP54 flatness risk. *Owner: Fang + Chase. Decision needed: this week.*
- ****Add active cabinet cooling.**** Spec a 400 W IP54 cabinet cooler or air-to-liquid heat exchanger tied to the main glycol loop. *Owner: Fang + Liquid Cooling module owner. Decision needed: within 2 weeks before cabinet panel cutouts are finalised.*
- ****Lock the EMS platform decision and dual-source the controller hardware.**** Pick CODESYS-compatible to keep portability. *Owner: Max (CTO) with Fang for manufacturing impact. Decision needed: within 2 weeks — this is on the critical path for the 16-week lead time.*
- ****Resolve SIL rating with insurer in writing.**** *Owner: founder + Leo. This week.*
- ****Specify ISO 2768-m for the cabinet, explicit tolerances for PCBA mounting and harness lengths.**** *Owner: Jian to add to drawings. Within 1 week.*
-

****Architecture review on CAN bus redundancy.**** Decide single vs dual CAN before slave BMS PCBA layout is frozen.

***Owner:** Max + Jian. Within 2 weeks.*

- ****ASSEMBLY: Sequence** — assemble late, after container shell, racks, and cable trays are in.** The cabinet mounts to the container wall in the PCS bay; doing it before the racks are in means you'll be working around it for everything else.
- ****ASSEMBLY: Fixtures** — wall-mount template jig (laser-cut steel plate matching cabinet rear-panel hole pattern) for marking and drilling the container wall mounting points. One-off cost £200, saves 30 minutes per build and prevents misalignment.**
- ****ASSEMBLY: Critical assembly tolerances** — cabinet rear-wall flatness against container wall (± 1 mm over 2 m) for IP54 seal; door hinge alignment (± 0.5 mm) for gasket compression; DIN-rail horizontal level (± 1 mm over cabinet width) for terminal block alignment.**
- ****ASSEMBLY: Test points after this module is assembled** — (a) IP54 leak test with smoke or pressure decay, (b) 24 VDC PSU output and UPS hold-up timer test (e30 min), (c) CAN bus continuity to all slave BMS positions (before racks installed — use a loopback test fixture), (d) Ethernet ring redundancy failover test, (e) hardwired safety I/O loop test to PCS trip and suppression release dummy loads, (f) earth bond test $< 0.1 \Omega$.**
- ****ASSEMBLY: Dependencies** — must be assembled after: (1) container shell with cable penetrations sealed, (2) cable tray installed in PCS bay, (3) container earth bonding complete. Must be assembled before: battery rack final commissioning (slave BMS units talk back to this module), PCS commissioning (interlocks here), fire panel commissioning (suppression release wired through this module's safety I/O).**

3.6 · Fire Detection & Suppression System

Module id: fire_safety_system

750.00 kg · 14 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

VESDA aspirating smoke detection, off-gas sensing, gaseous/aerosol suppression, and deflagration venting per NFCC and NFPA 855 requirements.

WHY IT MATTERS

Lithium battery thermal runaway is the dominant safety risk for a containerised BESS and the single largest factor in NFCC, insurer, and DNO acceptance. A compliant, UL 9540A-validated fire safety system is a gate condition for planning permission, insurance, and grid connection — without it the asset cannot be commissioned or insured, regardless of electrical performance.

DESCRIPTION

A fully integrated, container-internal fire safety package combining very-early-warning aspirating smoke detection (VESDA-E), distributed Li-ion off-gas sensing, a clean-agent gaseous suppression system (Novec 1230), in-rack pneumatic detection tubing, and passive deflagration venting in the roof. The system operates on a multi-stage logic: off-gas sensors trigger Stage 1 alarm and PCS shutdown before visible smoke; VESDA Stage 2 confirms smoke and pre-arms the suppression system; cross-zoned dual-knock confirmation triggers Novec discharge into the container free volume; if a deflagration occurs despite suppression, roof-mounted vent panels relieve overpressure to protect the shell and surroundings. All wetted parts use 316L stainless or brass; cylinders are seamless steel painted RAL 3000. The control panel is EN 54 certified and interfaces with the EMS/BMS via volt-free contacts and Modbus TCP, also driving an external strobe/sounder and remote alarm to the site fire panel and 24/7 monitoring centre.

KEY PARTS (7)

- VESDA-E VEP aspirating smoke detector (Xtralis) with 4-pipe sampling network, sensitivity 0.005–20% obs/m, covering both rack rows at ceiling level
- Li-ion off-gas detector array (e.g. Nexceris Li-ion Tamer 2nd Gen) — H₂/CO/VOC electrochemical sensors, 1 per rack pair, providing pre-smoke early warning
- Novec 1230 (FK-5-1-12) gaseous suppression skid — 2 × 180 L cylinders at 42 bar, manifolded with directional valves, sized for ~6.5% concentration in 80 m³ free volume per ISO 14520
- Deflagration vent panels (Fike or equivalent) — 2 × 1.0 m² roof-mounted, set pressure ~50 mbar, sized per NFPA 68 KG calculation for LFP off-gas
- Fire alarm control panel (Kentec Syncro AS or equivalent), EN 54-2/4 certified, with 24 V DC backup battery, dual-knock release logic and manual abort/release stations at door end
- High-pressure stainless steel (316L) discharge pipework with 360° ceiling nozzles (ISO 14520 listed), 1 nozzle per ~12 m² zone
- Heat-rated linear pneumatic detection tubing (Firetrace) routed through each battery rack as secondary in-rack detection

FAILURE MODES (4)

- False discharge of Novec agent due to single-sensor fault, causing extended downtime and ~£15k recharge cost
- Failure to detect off-gas early enough due to sensor placement or airflow short-circuiting from HVAC, allowing progression to thermal runaway
- Deflagration vent panels fail to open at design pressure (corrosion, paint blockage), leading to container shell rupture
- Suppression agent fails to control reignition — Novec/clean agents do not cool cells, and LFP runaway can re-propagate after agent dissipates

OPEN QUESTIONS (3)

- Whether the AHJ and site insurer will accept gaseous suppression alone or mandate water-mist/sprinkler backup per latest NFCC guidance
- UL 9540A large-scale fire test results for the chosen cell/module/rack combination — required to justify 3 m external separation distance
- Final free-volume calculation post-rack-installation, which determines exact agent quantity and cylinder count

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
VESDA-E VEP aspirating smoke detector — COTS aspirating detector frombuy Xtralis/supplier.	—		£2,000
Li-ion off-gas detector array (×6 sensors plus controller) — COTS Nexcerisbuy system for 6 zones, £1000 each.	—		£6,000
Novec 1230 suppression skid (2×180L cylinders, manifold, valves) — COTSbuy skid from fire suppression supplier, custom config.	—		£18,000
Deflagration vent panels (2×1.0 m² roof-mounted) — COTS Fike vent pan- make els, £1500 each.		Manual/Assembly · Other	£3,000
Fire alarm control panel (Kentec Syncro AS) — COTS panel with EN54 certmake from RS or supplier.		Manual/Assembly · Other	£1,500
High-pressure stainless steel discharge pipework with nozzles (×7 zones) make — Custom pipe fabrication per ISO 14520 layout.		Welding · Stainless Steel	£4,000
Heat-rated linear pneumatic detection tubing (Firetrace) ×12 — COTS Fire-buy trace tubing kits for each rack.	—		£1,800
Labour	£4,000 — 100 hrs at £40/hr for pipework, wiring, commissioning.		
Per-unit total	£40,300		

ASSUMPTIONS

- Suppression skid is COTS but needs custom manifolding.
- Discharge pipework is custom fabricated on-site.
- Prices exclude install labour for vent panels.

Mostly COTS suppression/detection items with custom 316L pipework.

4. BOM master (53 rows)

BOM derived from the module decomposition's keyParts, expanded into typed part rows. Part records live in the `parts` table and are joined back to modules via source_module_id.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
BAT-002	Battery Rack Steel Weldment — Welded steel rack frame providing structural support for stacked battery modules, with formed side panels, module support rails, cable management cutouts, and lifting/anchoring points.	Battery Rack Array (LFP)	make	sheet_metal · Carbon Steel · $\pm\pm1.5\text{mm}$	180.00kg	£650
BAT-005	Aluminium Liquid Cooling Cold Plate — CNC-machined aluminium liquid cooling cold plate with internal serpentine flow channels (FSW lid), providing uniform thermal control across the prismatic cell stack via 50/50 water-glycol coolant.	Battery Rack Array (LFP)	make	cnc · 6061 Aluminium · $\pm\pm0.1\text{mm}$	4.20kg	£220
BAT-007	Tin-Plated Copper Top-of-Rack Busbar — Tin-plated copper busbar mounted at the top of each battery rack to interconnect module terminals and carry rack-level DC current to the combiner. Sized for high continuous current with low IR drop.	Battery Rack Array (LFP)	make	machining · Electrolytic Copper · $\pm\pm0.2\text{mm}$	2.60kg	£55
BAT-ASY	Battery Rack Array Assembly — Top-level battery rack array assembly comprising multiple liquid-cooled battery modules, rack weldments, DC combiners, and slave BMS units, arranged in two longitudinal rows to deliver 3.5 MWh of DC storage.	Battery Rack Array (LFP)	make	other · Mixed Assembly · $\pm\pm2.0\text{mm}$	22000.00-kg	£420,000
BAT-MOD-ASY	Liquid-Cooled Battery Module — Liquid-cooled battery module sub-assembly containing a series/parallel arrangement of LFP prismatic cells, integrated cold plate, busbars, module enclosure, and slave BMS interface. Forms the field-replaceable unit within each rack.	Battery Rack Array (LFP)	make	other · Mixed Assembly · $\pm\pm1.0\text{mm}$	130.00kg	£1,850
BAT-PUR-001	LiFePO4 Prismatic Cell — Large-format prismatic LiFePO4 cell, 3.2V nominal ~280Ah, used as the base electrochemical building block for the battery modules. Aluminium hard case with laser-welded terminals.	Battery Rack Array (LFP)	buy	—	5.40kg	£75
BAT-PUR-003	Rack-Level DC Combiner with MCCB/Pyro-Fuse — Rack-level DC combiner box integrating molded-case circuit breaker, pyrotechnic fuse, current/voltage sensing, and busbar terminations for safe rack isolation under fault conditions.	Battery Rack Array (LFP)	buy	—	18.00kg	£1,450

BAT-PUR-004	Slave BMS Module (BMU) — Slave Battery Management Unit (BMU) monitoring per-cell voltages, module temperatures, and balancing, communicating to the master BMS via CAN/daisy-chain isoSPI link.	Battery Rack Array	buy	—	0.40kg	£180
BAT-PUR-006	Dripless Quick-Connect Coolant Coupling — Dripless quick-connect coolant coupling for liquid-cooled battery module manifold connections. Allows hot-swap of modules without coolant loss or air ingress into the glycol loop.	Battery Rack Array	buy	—	0.18kg	£45
BAT-PUR-008	Busbar Insulating Boot — Snap-on insulating boot covering exposed busbar joints and terminal lugs to provide IP2X finger protection and prevent inter-phase shorts. Flame-retardant elastomer rated for high-voltage DC service.	Battery Rack Array	buy	—	0.05kg	£5
BAT-PUR-009	Off-Gas Sensor Head — Early-warning off-gas sensor head detecting electrolyte vapor (H ₂ , CO, VOC, electrolyte carbonates) emitted from LFP cells before thermal runaway. Mounted within rack airflow path and wired to the fire detection controller.	Battery Rack Array	buy	—	0.25kg	£850
BESS-ASY	Containerised BESS Top-Level Assembly — Top-level assembly of the containerised 3.5 MWh / 1.5 MW BESS, integrating the ISO container shell, battery racks, PCS bay, thermal management, BMS/EMS controls, and fire safety system into a factory-tested transportable unit.	ISO Container Shell & Fire-Rated Enclosure	make	other · Mixed Assembly · ±±5mm overall envelope, ISO 1496 corner casting positions	38000.00kg	£850,000
CONT-001	40ft High Cube Corten Container Shell — Standard 40ft high-cube ISO shipping container shell in Corten weathering steel, used as the primary structural envelope. Includes side walls, end walls, doors, roof and base frame ready for fit-out.	ISO Container Shell & Fire-Rated Enclosure	buy	—	3900.00kg	£4,200
CONT-005	Reinforced Floor Unistrut Grid — Reinforced floor grid fabricated from hot-dip galvanised Unistrut/cold-formed C-channel, welded and bolted to the container base rails to support battery rack loads and distribute point loads to ISO corners.	ISO Container Shell & Fire-Rated Enclosure	make	sheet_metal · Galvanised Mild Steel · ±±2mm flatness over 12m, ±1mm hole pitch	380.00kg	£2,200
CONT-007	Roof Reinforcement Frame — Roof reinforcement frame fabricated from welded RHS and cold-formed channel to support the roof-mounted chiller, HVAC and deflagration vent loads, and to act as a crane-lift compatible substructure.	ISO Container Shell & Fire-Rated Enclosure	make	sheet_metal · Mild Steel · ±±2mm flatness, ±1mm bolt-hole pitch	220.00kg	£1,650

CONT-ASY	ISO Container Shell & Fire-Rated Enclosure Assembly — Top-level assembly comprising the 40ft high-cube ISO container shell with integrated fire-rated lining, deflagration vents, reinforced floor and roof, ISO corner castings, and cable transit frame. Provides the structural and fire-rated envelope for the BESS unit.	ISO Container Shell make & Fire-Rated Enclosure	other · Mixed Assembly · $\pm\pm 5\text{mm}$ overall (ISO 668)	4850.00kg	£28,500
CONT-PUR-002	ISO 1161 Forged Corner Casting — ISO 1161 compliant forged steel corner casting for crane lifting and ISO twist-lock anchoring. Eight units per container at the top and bottom corners.	ISO Container Shell buy & Fire-Rated Enclosure	—	8.50kg	£35
CONT-PUR-003	Calcium-Silicate Fire-Rated Lining Panel — Non-combustible calcium silicate fire-rated lining board mounted to interior walls and ceiling to provide A1 fire compartmentation per NFPA 855. Supplied as 1200x2400mm panels for site cutting and mechanical fix.	ISO Container Shell buy & Fire-Rated Enclosure	—	62.60kg	£145
CONT-PUR-004	Deflagration Vent Panel — Pressure-relief deflagration vent panel installed in container roof/end wall to release overpressure during a thermal runaway event per NFPA 68. Lightweight rupture panel with tethered retention.	ISO Container Shell buy & Fire-Rated Enclosure	—	18.00kg	£1,250
CONT-PUR-006	Multi-Cable Transit Gland Frame — Multi-cable transit (MCT) gland frame installed in the end wall providing IP66, fire-rated and EMC-bonded penetration for AC power, DC, control and fibre cabling to the external transformer and grid connection.	ISO Container Shell buy & Fire-Rated Enclosure	—	12.50kg	£850
EMS-006	Wall-Mounted Control Cabinet Enclosure — Wall-mounted IP54 control cabinet housing the EMS controller, network switch, UPS, and safety I/O. Hinged lockable door with mounting plate and cable entry glands; sized for 40 cm DIN rail mounting depth.	BMS & Energy Management Controller make	sheet_metal · Mild Steel Sheet · $\pm\pm 1.0\text{mm}$	28.00kg	£420
EMS-ASY	BMS & EMS Controller Cabinet Assembly — Wall-mounted IP54 control cabinet housing the master BMS controller, EMS industrial PC/PLC, communication gateways, 24VDC PSU, and terminal blocks; provides the central nervous system of the BESS.	BMS & Energy Management Controller make	other · Powder-coated mild steel enclosure with DIN rail interior · $\pm\pm 2\text{mm}$ assembly	55.00kg	£4,200
EMS-PUR-001	Master BMS Controller Unit — Master battery management system controller aggregating per-rack slave BMS units, performing SOC/SOH calculation, contactor control, and safety interlocks; communicates with EMS via Modbus TCP and CAN.	BMS & Energy Management Controller buy	—	1.80kg	£1,850

EMS-PUR-002	EMS Industrial PC / PLC — Industrial fan-less PC/PLC running the energy management application, grid-code compliance logic, SCADA gateway, and supervisory control of PCS, thermal, and fire subsystems.	BMS & Energy Management Controller	Man-buy	—	2.50kg	£1,650
EMS-PUR-003	Managed Ring-Redundant Ethernet Switch — Industrial managed Ethernet switch with ring redundancy (e.g. MRP/HSR) for resilient communication between BMS, PCS, EMS, and SCADA. DIN-rail mounted with 8x copper plus 2x SFP fibre uplinks.	BMS & Energy Management Controller	Man-buy	—	1.20kg	£950
EMS-PUR-004	24 VDC UPS-Backed Auxiliary Power Supply — 24 VDC uninterruptible auxiliary power supply with integrated battery backup providing ride-through power to the BMS, EMS controller, fire panel, and safety I/O during AC dropouts.	BMS & Energy Management Controller	Man-buy	—	6.50kg	£1,100
EMS-PUR-005	SIL 2 Hardwired Safety I/O Module — SIL 2 certified hardwired safety I/O module providing dedicated emergency stop, fire trip, and contactor open signals independent of the main EMS PLC. Fail-safe relay outputs to PCS and DC contactors.	BMS & Energy Management Controller	Man-buy	—	0.90kg	£1,450
FAS-PUR-001	M30 Grade 8.8 Chemical Foundation Anchor — M30 chemical/resin-bonded foundation anchor stud used to secure the ISO container corner castings and base rails to the prepared concrete pad. Supplied with hex nut, washer, and resin capsule.	ISO Container Shell & Fire-Rated Enclosure	buy	—	0.85kg	£13
FIR-005	Stainless Steel Suppression Discharge Pipework — Stainless steel discharge pipework distributing gaseous/aerosol suppression agent from the cylinder bank to ceiling nozzles throughout the container. Machined, threaded, and welded sections sized per NFPA 2001 hydraulic calculations.	Fire Detection & Suppression System	make	machining · 316 Stainless Steel · $\pm\pm 0.5\text{mm}$ on lengths, $\pm 0.2\text{mm}$ on threads	22.00kg	£780
FIR-ASY	Fire Detection & Suppression System Assembly — Top-level assembly of the fire detection and suppression system, integrating aspirating smoke detection, off-gas sensors, Novec 1230 cylinder skid, pipework, nozzles, and fire control panel within the container.	Fire Detection & Suppression System	make	other · Mixed Assembly · $\pm\text{N/A}$ (assembly)	220.00kg	£28,500
FIR-PUR-001	VESDA-E Aspirating Smoke Detector — Xtralis VESDA-E VEU aspirating smoke detector with sampling pipe network drawing air from each battery rack aisle, providing very early warning smoke detection at four programmable alarm thresholds.	Fire Detection & Suppression System	buy	—	4.50kg	£4,200

FIR-PUR-002	Li-ion Off-Gas Detector Array — Array of Li-ion battery off-gas detectors (e.g. Honeywell Li-ion Tamer or Nexceris) providing earliest warning of cell venting by detecting electrolyte vapours before smoke is produced. One sensor per rack zone with controller.	Fire Detection & Suppression System	buy	—	3.50kg	£6,800
FIR-PUR-003	Novec 1230 Suppression Cylinder Skid — Pre-engineered Novec 1230 (FK-5-1-12) clean agent suppression cylinder skid with 180L cylinder, discharge valve, pressure switch, manifold, and mounting frame. Sized for total flooding of the container free volume.	Fire Detection & Suppression System	buy	—	280.00kg	£14,500
FIR-PUR-004	Fire Alarm Control Panel — UL-listed fire alarm control panel coordinating VESDA aspirating detectors, off-gas sensors, suppression release, and audible/visual alarms. Provides addressable loop inputs, releasing circuits, and Modbus/IP interface to the BESS supervisory controller.	Fire Detection & Suppression System	buy	—	9.50kg	£1,850
FIR-PUR-006	Ceiling Discharge Nozzle — Brass ceiling-mounted discharge nozzle with calibrated orifice for uniform agent distribution over the battery rack aisles. UL/FM listed for clean-agent or aerosol suppression service.	Fire Detection & Suppression System	buy	—	0.18kg	£38
FIR-PUR-007	Pneumatic Linear Heat Detection Tubing — Pneumatic linear heat detection tubing routed along battery rack tops; ruptures at a calibrated temperature to release suppression agent directly at the source. Provides redundant detection independent of the electronic FACP.	Fire Detection & Suppression System	buy	—	1.80kg	£320
PCS-004	Cable Gland Plate — Sheet metal gland plate sealing the PCS bay's external cable entry point. Provides IP54-rated penetrations for AC power cables to the external transformer and auxiliary control cabling.	Power Conversion System Bay	make	sheet_metal · Mild Steel · ±±0.5mm	8.50kg	£145
PCS-005	Internal Liquid Cooling Manifold — Machined aluminium liquid cooling manifold distributing coolant to PCS power module cold plates. Includes inlet/outlet ports, internal flow channels, and mounting bosses for IGBT stack cooling.	Power Conversion System Bay	make	machining · 6061 Aluminium · ±±0.1mm	6.20kg	£380
PCS-007	Bay Segregation Fire-Rated Partition — Fire-rated sheet metal partition segregating the PCS bay from adjacent battery racks. Provides 60-minute fire integrity per EN 1364 with intumescent seals at edges.	Power Conversion System Bay	make	sheet_metal · Galvanised Steel with Mineral Wool Core · ±±1mm	65.00kg	£880

PCS-ASY	Power Conversion System Bay Assembly — Top-level assembly of the Power Conversion System bay, integrating the bi-directional inverter stack, AC switchgear, DC protection panel, and bay-level wiring/ducting within the container's PCS compartment.	Power Conversion System Bay	make	other · Mixed assembly · ±±2mm	1850.00kg	£145,000
PCS-PUR-001	Bi-Directional Grid-Forming PCS Inverter Stack — Bi-directional grid-forming PCS inverter stack rated 1.5 MW, providing four-quadrant power conversion between the LFP battery DC bus and 400V AC. Supports virtual synchronous machine control, black-start, and grid-code compliance.	Power Conversion System Bay	buy	—	1200.00kg	£95,000
PCS-PUR-002	AC Switchgear Cabinet with MCCB — Low-voltage AC switchgear cabinet housing the main MCCB, metering, surge protection, and grid-tie disconnect for the 1.5 MW PCS output. Provides isolation and short-circuit protection between PCS AC bus and external transformer.	Power Conversion System Bay	buy	—	320.00kg	£18,500
PCS-PUR-003	DC Isolation & Protection Panel — Isolation and protection panel between battery rack array and PCS DC input, containing motorized DC contactors, fuses, pre-charge circuit, insulation monitoring, and ground-fault detection.	Power Conversion System Bay	buy	—	280.00kg	£15,500
PCS-PUR-006	Auxiliary LV Distribution Panel — Auxiliary low-voltage distribution panel providing 400/230VAC supply to ancillary loads (HVAC, controls, lighting, BMS). Includes MCBs, RCDs, and surge protection in a wall-mount enclosure.	Power Conversion System Bay	buy	—	12.00kg	£650
THM-003	Stainless Steel Coolant Ring Main — Stainless steel ring main piping distributing coolant along both battery rack rows with branch tees to each rack. Welded and flanged construction with insulation jacket to prevent condensation.	Liquid Cooling & HVAC System	make	machining · 304 Stainless Steel · ±±1mm	95.00kg	£2,400
THM-ASY	Liquid Cooling & HVAC System Assembly — Top-level assembly of the thermal management system including chiller, CDU, ring main piping, HVAC, valves, sensors, and insulation. Integrates all coolant and air-conditioning components for the BESS.	Liquid Cooling & HVAC System	make	other · Mixed Assembly · ±N/A	850.00kg	£42,000

THM-PUR-001	Roof-Mounted Packaged Chiller — Roof-mounted packaged air-cooled chiller providing approximately 80kW cooling capacity for the BESS coolant loop. Includes scroll compressors, condenser fans, and integrated controls with Modbus interface.	Liquid Cooling & HVAC System	buy	—	720.00kg	£18,500
THM-PUR-002	Coolant Distribution Unit (CDU) — Coolant Distribution Unit providing pumps, expansion tank, filtration, and heat exchanger interface between primary chiller loop and secondary battery rack loop. Redundant pumps with VFD control.	Liquid Cooling & HVAC System	buy	—	280.00kg	£9,800
THM-PUR-004	EPDM Flexible Coolant Jumper — Flexible EPDM rubber jumper hose with stainless braid and quick-disconnect fittings for connecting rigid coolant manifolds to battery rack cold plates while accommodating thermal expansion and vibration.	Liquid Cooling & HVAC System	buy	—	0.85kg	£38
THM-PUR-005	Internal HVAC Unit — Wall-mounted internal HVAC split unit providing dehumidification and ambient air conditioning of the container interior to maintain enclosure dew point and electronics temperature within spec.	Liquid Cooling & HVAC System	buy	—	42.00kg	£1,450
THM-PUR-006	PT1000 Temperature Sensor — Class A PT1000 RTD temperature sensor with M12 connector and stainless probe sheath for monitoring coolant supply/return and ambient temperatures across racks and the HVAC loop.	Liquid Cooling & HVAC System	buy	—	0.08kg	£32
THM-PUR-007	Ultrasonic Coolant Flow Meter — Inline clamp-on ultrasonic flow meter for non-invasive measurement of glycol coolant flow rate in the primary distribution loop, providing 4-20mA output to the EMS for thermal balance monitoring.	Liquid Cooling & HVAC System	buy	—	1.20kg	£685
THM-PUR-008	Glycol Leak Detection Rope Sensor — Length-locating glycol leak detection rope sensor routed along coolant manifold drip trays; provides early warning and pinpoints leak location to the BMS/fire panel.	Liquid Cooling & HVAC System	buy	—	0.45kg	£240

5. Cost waterfall

Cost estimates grounded against the material-properties library (47 rows, freshest 23 Mar 2026) and the `process\_capabilities` table (20 rows, freshest unknown). Costs sit in the project's ai_cost_estimates jsonb, keyed by module id.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£1,276,480	£180,000	£-1,096,480

Per-module roll-up

MODULE	COST	% OF UNIT
ISO Container Shell & Fire-Rated Enclosure	£21,880	1.7%
Battery Rack Array (LFP)	£817,600	64.1%
Power Conversion System Bay	£316,900	24.8%
Liquid Cooling & HVAC System	£52,300	4.1%
BMS & Energy Management Controller	£27,500	2.2%
Fire Detection & Suppression System	£40,300	3.2%

6. Risks register

Every failure mode and open question declared against each module, in one register. Until a dedicated risks store ships with severity + ownership, these rows are the canonical risk inventory.

ISO Container Shell & Fire-Rated Enclosure

KNOWN FAILURE MODES (4)

- Deflagration vent under-sizing or sticking, causing overpressure rupture of side walls during a thermal-runaway off-gas ignition event
- Corrosion of Corten shell at cable-gland penetrations and roof-vent flashings, leading to water ingress onto LV busbars and DC arc faults
- Fatigue cracking at corner-casting welds from road transport vibration combined with high payload (45–55 t), risking lifting failure on site
- Fire-rated lining degradation or thermal bridging at penetrations, reducing compartment rating below the 120 min specified and breaching NFCC separation distances

OPEN QUESTIONS (3)

- Confirmed gross transport mass — at the upper end (55 t) the assembly exceeds standard UK 44 t GVW and requires STGO Cat 2 abnormal-load permitting
- Required deflagration vent area, dependent on UL 9540A large-scale fire test results for the chosen cell and module geometry
- Whether AHJ and insurer will accept a single-compartment shell or mandate internal fire-rated segregation between battery zone and PCS bay

Battery Rack Array (LFP)

KNOWN FAILURE MODES (4)

- Single-cell thermal runaway propagating module-to-module due to inadequate cold-plate contact or insufficient inter-module fire barriers
- DC busbar joint loosening from thermal cycling causing localised I^2R hotspots and arc-flash risk
- Coolant leak at quick-disconnect couplings causing DC insulation fault and rack trip
- Cell capacity divergence (top-balancing drift) reducing usable energy below 3.5 MWh contractual threshold within 2–3 years

OPEN QUESTIONS (3)

- Final cell supplier and exact cell count — drives rack quantity, total mass, and whether 44 t road limit is achievable
- Rack DC voltage architecture (1500 V single string vs. lower-voltage paralleled) — affects PCS selection and isolation strategy
- Required UL 9540A test configuration (unit-level vs. installation-level) and whether inter-rack fire barriers are needed to pass

Power Conversion System Bay

KNOWN FAILURE MODES (4)

- IGBT/SiC module failure due to thermal cycling or coolant flow loss, causing inverter trip and loss of dispatch
- DC bus insulation degradation or earth fault leading to IMR alarm and forced shutdown
- AC breaker fails to clear a downstream fault, causing arc-flash event within the bay
- Coolant leak from manifold or cold plate onto live DC busbars, risking short-circuit and fire propagation

OPEN QUESTIONS (3)

- Final PCS vendor selection (e.g. Sungrow, SMA, Power Electronics, EPC) and whether 4× 375 kW or 6× 250 kW topology is preferred for redundancy vs. cost
- Whether the LV/MV transformer is supplied within the container envelope or as an external skid — affects gland plate sizing and AC voltage class
- Confirmation that grid-forming firmware meets the specific DNO's interpretation of G99 / G100 at the time of commissioning

Liquid Cooling & HVAC System

KNOWN FAILURE MODES (4)

- Glycol leak at rack jumper coupling causing low-flow alarm, BMS derate, and potential DC arc-fault if coolant contacts busbars
- Compressor failure or refrigerant loss in chiller leading to total loss of battery cooling and forced system shutdown
- Pump bearing wear or controller fault on duty pump without successful failover to standby, causing localised hot-spots
- Condensation on cold coolant lines or cold plates in humid UK summer conditions, dripping onto electrical equipment

OPEN QUESTIONS (3)

- Final heat load split between battery racks (continuous ~60–80 kW) and PCS (peak ~30–45 kW) pending PCS efficiency curve confirmation
- Whether a single chiller is acceptable or N+1 redundancy is required by insurer / customer SLA
- Acoustic emission limits at site boundary — may drive condenser fan selection and night-mode operation logic

BMS & Energy Management Controller

KNOWN FAILURE MODES (4)

- Loss of CAN communication to one or more slave BMS modules causing loss of cell visibility and forced derate or trip
- EMS firmware fault or watchdog hang leading to stale dispatch commands and possible grid-code non-compliance
- Auxiliary 24 VDC supply failure with UPS depletion, causing uncontrolled shutdown of contactors and cooling
- Cyber-intrusion via SCADA/remote-access interface enabling unauthorised dispatch or safety-system bypass

OPEN QUESTIONS (3)

- Final selection of EMS platform and whether grid-forming control logic is hosted on EMS PLC or embedded in PCS firmware
- Required SIL rating for the suppression-release and contactor-trip safety function (SIL 2 vs SIL 3) per insurer/AHJ
- Northbound protocol stack required by target DNOs and aggregator partners (IEC 61850 vs DNP3 vs proprietary)

Fire Detection & Suppression System

KNOWN FAILURE MODES (4)

- False discharge of Novec agent due to single-sensor fault, causing extended downtime and ~£15k recharge cost
- Failure to detect off-gas early enough due to sensor placement or airflow short-circuiting from HVAC, allowing progression to thermal runaway
- Deflagration vent panels fail to open at design pressure (corrosion, paint blockage), leading to container shell rupture
- Suppression agent fails to control reignition — Novec/clean agents do not cool cells, and LFP runaway can re-propagate after agent dissipates

OPEN QUESTIONS (3)

- Whether the AHJ and site insurer will accept gaseous suppression alone or mandate water-mist/sprinkler backup per latest NFCC guidance
- UL 9540A large-scale fire test results for the chosen cell/module/rack combination — required to justify 3 m external separation distance
- Final free-volume calculation post-rack-installation, which determines exact agent quantity and cylinder count

7. Supplier shortlist (29)

Shortlist built by scoring each supplier in the directory (651 companies) and marketplace listings (28,315 listings) against each module's declared process + material. A low shortlist count usually means the directory doesn't yet have coverage for the project's niche — not that no match exists globally.

amberpanels.co.uk

Amber Panels

<https://www.amberpanels.co.uk/>

Contact: enquiries@amberpanels.co.uk

HQ not declared · ramp role unassigned · match score 34.8

Matched across 1 module: container_enclosure

SCORE BREAKDOWN

total	34.8
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	21.4
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Industry: Marine
- Products

eldapoint-group.co.uk

Eldapoint Group

<https://www.eldapoint-group.co.uk/sectors/renewables/>

HQ not declared · ramp role unassigned · match score 30.8

Matched across 1 module: container_enclosure

SCORE BREAKDOWN

total	30.8
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20.4
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Marine
- Products

cfstruthers.co.uk

CFS

<https://www.cfstruthers.co.uk/services-and-installation>

Contact: sales@cfstruthers.co.uk

HQ not declared · ramp role unassigned · match score 30.7

Matched across 1 module: container_enclosure

SCORE BREAKDOWN

total	30.7
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Marine
- Heavy Fabrication & Pressure Vessels

clydepc.co.uk

Clyde Pneumatic Conveying

<https://clydepc.co.uk/>

Contact: sales@clydepc.co.uk

HQ not declared · ramp role unassigned · match score 23.4

Matched across 1 module: Fire Detection & Suppression System

SCORE BREAKDOWN

total	23.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	20
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Products

atscontrols.co.uk

ATS Controls

<https://atscontrols.co.uk/custom-control-panel-manufacturer-for-every-industry/>

Contact: sales@atscontrols.co.uk

HQ not declared · ramp role unassigned · match score 31.4

Matched across 1 module: BMS & Energy Management Controller

SCORE BREAKDOWN

total	31.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

nidec.com

Coolant Distribution Units (CDU = water-cooling system) to respond to growing thermal loads

<https://www.nidec.com/en/technology/casestudy/liquid-cooling/>

HQ not declared · ramp role unassigned · match score 30.8

Matched across 1 module: Liquid Cooling & HVAC System

SCORE BREAKDOWN

total	30.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	30.4
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

gpsgroup.co.uk

RSE Control Systems

<https://gpsgroup.co.uk/>

Contact: sales.gps@onerse.com

HQ not declared · ramp role unassigned · match score 31.3

Matched across 1 module: control_ems_cabinet

SCORE BREAKDOWN

total	31.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

huaxiangbattery.com

Supply LITHIUM IRON PHOSPHATE BATTERY RACK ENERGY STORAGE MODULE Wholesale Factory

<https://www.huaxiangbattery.com/product/lithium-iron-phosphate-battery-rack-energy-storage-module>

Contact: sales22@huaxiangbattery.com

HQ not declared · ramp role unassigned · match score 33.4

Matched across 1 module: Battery Rack Array (LFP)

SCORE BREAKDOWN

total	33.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	33
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

trumony.com

Liquid Cooling Unit for Battery Energy Storage System (BESS) Rack

<https://www.trumony.com/Liquid-Cooling-Unit-for-Battery-Energy-Storage-System-BESS-Rack-pd42291769.html>

Contact: tracy@trumony.com

HQ not declared · ramp role unassigned · match score 26.9

Matched across 1 module: ISO Container Shell & Fire-Rated Enclosure

SCORE BREAKDOWN

total	26.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

keswitchgear.co.uk

G99 Protection Panels

<https://www.keswitchgear.co.uk/renewables/g99-protection-panels/>

Contact: info@keswitchgear.co.uk

HQ not declared · ramp role unassigned · match score 31.4

Matched across 1 module: control_ems_cabinet

SCORE BREAKDOWN

total	31.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

symtechsolar.com

373kWh Liquid Cooled Energy Storage System

<https://www.symtechsolar.com/battery-energy-storage-systems/373kw-liquid-cooled-battery-energy-storage-system/>

Contact:

HQ not declared · ramp role unassigned · match score 33.6

Matched across 1 module: Battery Rack Array (LFP)

SCORE BREAKDOWN

total	33.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	33.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

cmxbattery.com

LiFePo4 19" Rack-Mount Li-Ion Lithium iron phosphate server rack Battery

<https://cmxbattery.com/Product/19-rack-mount-li-ion-battery-with-lithium-iron-phosphate-technology/>

Contact: sales@cmxbattery.com

HQ not declared · ramp role unassigned · match score 33.6

Matched across 1 module: Battery Rack Array (LFP)

SCORE BREAKDOWN

total	33.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	33.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

guchen.com

Battery Chiller

<https://www.guchen.com/battery-solutions/ev-battery-chiller.html>

Contact: info@guchen.com

HQ not declared · ramp role unassigned · match score 31.7

Matched across 1 module: Liquid Cooling & HVAC System

SCORE BREAKDOWN

total	31.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	31.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

scupower.com

PCS Power Conversion System Energy Storage, PCS Converter/Hybrid Inverter System

<https://www.scupower.com/energy-storage/power-conversion-system/>

Contact: enquiry@scupower.com

HQ not declared · ramp role unassigned · match score 33.6

Matched across 1 module: Power Conversion System Bay

SCORE BREAKDOWN

total	33.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	33.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

sens-usa.com

Switchgear Power Systems

<https://www.sens-usa.com/applications/switchgear>

Contact: info@sens-usa.com

HQ not declared · ramp role unassigned · match score 31.0

Matched across 1 module: pcs_cabinet

SCORE BREAKDOWN

total	31
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.6
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

elecod.com

50kW-500kW C&I Power Conversion System (PCS)

<https://www.elecod.com/product/monet-series-50kw-500kw-c-i-power-conversion-system-pcs>

Contact: info@en.elecod.com

HQ not declared · ramp role unassigned · match score 32.7

Matched across 1 module: Power Conversion System Bay

SCORE BREAKDOWN

total	32.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	32.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

pyrophobic.com

Guide to Thermal Runaway Mitigation and Containment

<https://pyrophobic.com/blog/thermal-runaway-mitigation-and-containment/>

HQ not declared · ramp role unassigned · match score 25.6

Matched across 1 module: Fire Detection & Suppression System

SCORE BREAKDOWN

total	25.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	22.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturing

accupanel.com

DC DISTRIBUTION BOARD

<https://www.accupanel.com/dc-distribution-board/>

HQ not declared · ramp role unassigned · match score 32.6

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	32.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

getfse.com

FSE

<https://getfse.com/control-panels>

Contact: sales@getfse.com

HQ not declared · ramp role unassigned · match score 25.3

Matched across 1 module: Fire Detection & Suppression System

SCORE BREAKDOWN

total	25.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Products

engineersguidebook.com

Air Circuit Breakers (ACB)

<https://engineersguidebook.com/air-circuit-breakers-acb/>

HQ not declared · ramp role unassigned · match score 30.1

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	30.1
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	29.7
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

vectorenergy.com

MCCB DC Molded Case Circuit Breakers

<https://vectorenergy.com/en/product/interruptores-en-caja-moldeada-mccb-dc/>

HQ not declared · ramp role unassigned · match score 31.3

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	31.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

hitachienergy.com

Battery Power Conversion System (PCS)

<https://www.hitachienergy.com/products-and-solutions/power-conversion/battery-power-conversion-system-pcs>

HQ not declared · ramp role unassigned · match score 34.2

Matched across 1 module: Power Conversion System Bay

SCORE BREAKDOWN

total	34.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	33.8
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

cooltec.de

COOLTEC

<https://cooltec.de/en/>

Contact: info@cooltec.de

HQ not declared · ramp role unassigned · match score 37.5

Matched across 1 module: Liquid Cooling & HVAC System

SCORE BREAKDOWN

total	37.5
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	27.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Thermal Management & Cooling Systems

SCORE BREAKDOWN

total	27.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.2
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Products

nammo.com

Nammo UK Ltd (Nammo Westcott)

<https://www.nammo.com/location/westcott/>

Contact: sales@nammo.com

HQ not declared · ramp role unassigned · match score 20.4

Matched across 1 module: Liquid Cooling & HVAC System

SCORE BREAKDOWN

total	20.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	0
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Hardware Company

compact-kaeltetechnik.de

compact Kältetechnik GmbH

<https://www.compact-kaeltetechnik.de/en/>

HQ not declared · ramp role unassigned · match score 30.7

Matched across 1 module: Liquid Cooling & HVAC System

SCORE BREAKDOWN

total	30.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Refrigeration Systems

isoclad.co.uk

Isoclad

<https://www.isoclad.co.uk/company-overview/>

Contact: info@isoclad.co.uk

HQ not declared · ramp role unassigned · match score 26.8

Matched across 1 module: ISO Container Shell & Fire-Rated Enclosure

SCORE BREAKDOWN

total	26.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.4
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Insulated Panels & Cladding Systems

maxwellenergy.co

Battery Management System & BMS for High Power Systems

<https://www.maxwellenergy.co/blog/battery-management-system-bms-for-high-power-systems-the-complete-guide/>

HQ not declared · ramp role unassigned · match score 32.0

Matched across 1 module: BMS & Energy Management Controller

SCORE BREAKDOWN

total	32
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	31.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

integratedbms.co.uk

Integrated BMS Ltd

<https://www.integratedbms.co.uk/>

Contact: controls@integratedbms.co.uk

HQ not declared · ramp role unassigned · match score 31.2

Matched across 1 module: BMS & Energy Management Controller

SCORE BREAKDOWN

total	31.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing